

STUDENT NUMBER.....



GOSFORD HIGH SCHOOL

2020

HIGHER SCHOOL CERTIFICATE

FINAL ASSESSMENT

Year 11 PRELIMINARY MATHEMATICS – EXTENSION 1

Time Allowed - 90 minutes plus 10 minutes reading time

- Write using a black or blue pen. Black pen is preferred.
- Board approved calculators may be used
- Answers to the multiple choice are to be done on the answer sheet provided.
- Start each section on a **new** page. **Each section** will be collected **separately.**
- Show all relevant mathematical reasoning and/or calculations.
- **THIS PAPER IS TO BE RE-COLLECTED AT END OF EXAM.**

Section I

Attempt Questions 1-8 (8 Marks)

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for questions 1-10.

- 1 What is the Cartesian equation of the curve with parametric equations $x = 20t$ and $y = 40t - 5t^2$?

(A) $y = 2x - \frac{x^2}{400}$

(B) $y = 2x - \frac{x^2}{200}$

(C) $y = 2x - \frac{x^2}{160}$

(D) $y = 2x - \frac{x^2}{80}$

- 2 What is the domain of the function $f(x) = \ln(3x - x^2)$?

1

(A) $(0, 3)$

(B) $[0, 3)$

(C) $(0, 3]$

(D) $[0, 3]$

- 3 For what value of k is the polynomial $P(x) = kx^2 + (2k-1)x + k$ a perfect square?

1

(A) $k = \frac{1}{9}$

(B) $k = \frac{1}{4}$

(C) $k = 4$

(D) $k = 9$

- 4 Which of the following is NOT a factor of $P(x) = x^3 + 2x^2 - 5x - 6$?

1

(A) $(x+1)$

(B) $(x-2)$

(C) $(x-3)$

(D) $(x+3)$

Marks

- 5 What is the value of $\cos^{-1}(\tan x) + \sin^{-1}(\tan x)$ for $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$? 1
- (A) 0
(B) 1
(C) $\frac{\pi}{2}$
(D) π
- 6 Which of the following is NOT an expression for $\tan\left(\frac{x}{2} + \frac{\pi}{4}\right)$? 1
- (A) $\sec x + \tan x$
(B) $\frac{1 + \sin x}{\cos x}$
(C) $\frac{\cos x}{1 - \sin x}$
(D) $(1 + \tan x)^2$
- 7 In how many ways can a group of 6 people line up in a queue so that the oldest person is in front of the youngest person? 1
- (A) 180
(B) 360
(C) 540
(D) 720
- 8 What is the expansion of $(3 - 2\sqrt{2})^3$? 1
- (A) $27 - 16\sqrt{2}$
(B) $51 - 34\sqrt{2}$
(C) $99 - 70\sqrt{2}$
(D) $171 - 124\sqrt{2}$

Section II

Attempt Questions 9-12**Allow about 1 hour and 45 minutes for this section.**

Answer the questions in writing booklets provided. Use a separate writing booklet for each question. In Questions 11-14 your responses should include relevant mathematical reasoning and/or calculations.

Question 9 (12 Marks)**Use a separate writing booklet.**

- (a) Solve the inequality $|2x + 1| \geq 3$. 2
- (b) The equation $P(x) = 0$ where $P(x) = x^3 + cx + d$ has a double root $x = 1$. Find the values of c and d . 2
- (c) Show that $\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x} = \tan x$. 2
- (d) Solve the inequality $x + \frac{4}{x} \geq 5$. 3
- (e) Consider the polynomial $P(x) = k^2x^3 - k^3x^2 - x + k$ where $k \neq 0$.
- (i) Show that $(x - k)$ is a factor of $P(x)$. 1
- (ii) Solve the equation $P(x) = 0$. 2

Marks**Question 10 (12 Marks)****Use a separate writing booklet.**

(a) Find the number of ways that 4 students can be chosen from 6 juniors and 3 seniors

(i) without restriction. **1**

(ii) so that there are unequal numbers of juniors and seniors. **2**

(b) The equation $2x^3 + 6x^2 + 4x + 1 = 0$ has roots α , β and γ .

(i) Find the value of $(1 - \alpha) + (1 - \beta) + (1 - \gamma)$. **1**

(ii) Find the value of $(1 - \alpha)(1 - \beta)(1 - \gamma)$. **2**

(c) Write $\cos x - 3 \sin x + 3$ using the “ t ” results given that $t = \tan \frac{x}{2}$. **3**

d) **1**
1. Write down the binomial expansion of $(1 + x)^n$

11. Use the expansion of $(1 + x)^n$ to show that ${}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n$ **2**

Marks**Question 11 (12 Marks)****Use a separate writing booklet.**

- (a) Find the multiplicity of the root $x = -1$ of the equation $P(x) = 0$ where $P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$. 2
- (b) Sketch the graph of the function $f(x) = -2 \tan^{-1} x$ showing the intercepts on the axes and the equations of the asymptotes. 2
- (c) Solve the equation $\sin^2 \theta = \cos \theta + 1$ where $0 \leq \theta \leq 2\pi$. 3
- (d) Find the centre and radius of the circle $x^2 + y^2 - 6x + 2y + 6 = 0$ 2
- (e) Consider the function $f(x) = (x - 2)^2$, $0 \leq x \leq 2$.
Find the inverse function $f^{-1}(x)$ and state its domain. 3

Question 12 (12 Marks)

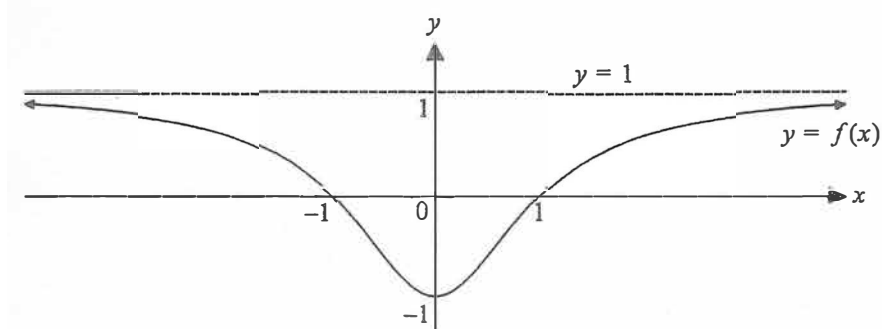
Use a separate writing booklet.

- (a) Find the number of ways in which the letters of the word TRIANGLE can be arranged in a line
- (i) so that all the vowels are next to each other and all the consonants are next to each other. 1
- (ii) so that consonants are in the two end positions. 1

- (b) If $\sec x + \tan x = k$, find in terms of k the value of $\sec x$. 2

- c) Find the number of ways in which 2 boys and 5 girls can be arranged in a circle
- i. if there is no restrictions 1.
- ii. if the boys are to sit together 1.

(d)



The diagram shows the graph of the function $f(x) = \frac{x^2 - 1}{x^2 + 1}$. On separate diagrams sketch the graphs of the following curves, showing clearly the intercepts on the axes and the equations of any asymptotes.

- (i) $y = |f(x)|$. 2
- (ii) $y = \frac{1}{f(x)}$. 2

- (e) i. Express $\cos x - \sqrt{3}\sin x$ in the form $R\cos(x + \alpha)$ where $R > 0$ and $0 \leq x \leq 2\pi$ 2

(1/8)

ANSWERS TO Y11 EXT ONE PRELIM FINAL 2020

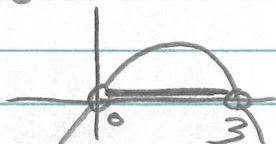
$$\begin{aligned} 1/ \quad x &= 20t \Rightarrow t = \frac{x}{20} \\ y &= 40t - 5t^2 \\ y &= 40\left(\frac{x}{20}\right) - 5\left(\frac{x^2}{20}\right) \end{aligned}$$

$$\begin{aligned} &= 2x - \frac{5x^2}{400} \\ &= 2x - \frac{x^2}{80} \end{aligned}$$

(D)

$$2/ \quad f(x) = \ln(3x - x^2)$$

for $3x - x^2 > 0$



$$\therefore \text{d: } 0 < x < 3$$

ie $(0, 3)$

(A)

$$3/ \quad P(x) = k(x^2 + (2 - \frac{1}{k})x + 1)$$

for $x^2 + (2 - \frac{1}{k})x + 1$
to be a perfect square

$$2 - \frac{1}{k} = -2$$

$$k = \frac{1}{4}$$

(B)

$$4/ \quad P(x) = x^3 + 2x^2 - 5x - 6$$
$$P(-1) = 0$$
$$P(2) = 0$$
$$P(3) = 24 \quad \checkmark (x-3)$$

check $P(-3) = 0$

(C)

$$5/ \quad \text{for } -\frac{\pi}{4} < x < \frac{\pi}{4}$$

$$-1 < \tan x < 1$$

$$\text{let } \tan x = u$$

$$\therefore \cos^{-1} u + \sin^{-1} u = \frac{\pi}{2}$$

for all $-1 < u < 1$

$\therefore$ (C)

6/ Using "t" results

$$\tan\left(\frac{x}{2} + \frac{\pi}{4}\right) = \frac{\tan \frac{x}{2} + \tan \frac{\pi}{4}}{1 - \tan \frac{x}{2} \tan \frac{\pi}{4}}$$

$$= \frac{t + 1}{1 - t}$$

$$A) \sec x + \tan x = \frac{1 + t^2 + 2t}{1 - t^2}$$

$$= \frac{(1+t)^2}{(1+t)(1-t)} = \frac{1+t}{1-t}$$

$$B) \frac{1 + \sin x}{\cos x} = \sec x + \tan x = A$$

$$C) \frac{\cos x}{1 - \sin x} = \frac{\cos x (1 + \sin x)}{(1 - \sin^2 x)}$$
$$= \frac{\cos x (1 + \sin x)}{\cos^2 x}$$

$$= \frac{1 + \sin x}{\cos x}$$

$$= \sec x + \tan x = A$$

$$D) x = -\frac{\pi}{4} \Rightarrow (1 + \tan x)^2 = 0$$

But

$$\tan\left(-\frac{\pi}{8} + \frac{\pi}{4}\right) = \tan \frac{\pi}{8} > 0$$

$\therefore$ NOT A (D)

$\frac{2}{8}$

7 older person
in front of younger

$$\frac{1}{2} \times 6! = 360$$

(B)

$$8(3 - 2\sqrt{2})^3$$

$$= 3^3 - 3 \times 3^2 \times 2\sqrt{2} + 3 \times 3 \times (2\sqrt{2})^2 - (2\sqrt{2})^3$$

$$= 27 - 54\sqrt{2} + 72 - 16\sqrt{2}$$

$$= 99 - 70\sqrt{2}$$

(C)

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Section II

Q9 a) $|2x+1| \geq 3$

$$2x+1 \leq -3 \quad 2x+1 \geq 3 \quad (1)$$

$$2x \leq -4 \quad 2x \geq 2$$

$$x \leq -2 \quad x \geq 1 \quad (1)$$

b) $P(x) = x^3 + cx + d$

roots at $x=1, 1$ & α

$$1+1+\alpha = -\frac{b}{a} = 0$$

$$\therefore \alpha = -2$$

$$\text{from } \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = c$$

$$-2 \times 1 + -2 \times 1 + 1 \times 1 = -3$$

$$c = -3$$

$$\text{from } \alpha\beta\gamma = -\frac{d}{a} = -d$$

$$d = -(1 \times 1 \times -2) = 2$$

$$\therefore c = -3$$

$$d = 2$$

c) $\frac{1 - \cos 2x + \sin 2x}{1 + \cos 2x + \sin 2x}$

$$= \frac{2\sin^2 x + 2\sin x \cos x}{2\cos^2 x + 2\sin x \cos x} \quad (1)$$

$$= \frac{2\sin x (\sin x + \cos x)}{2\cos x (\cos x + \sin x)} \quad (1)$$

$$= \tan x$$

d) $x + \frac{4}{x} \geq 5 \quad (x \neq 0)$

$$x^2 + 4x \geq 5x^2 \quad (1)$$

$$x^2 - 5x^2 + 4x \geq 0$$

$$x(x^2 - 5x + 4) \geq 0$$

$$x(x-4)(x-1) \geq 0 \quad (1)$$



(1)

$$0 < x \leq 1 \text{ and } x \geq 4$$

e) $P(x) = k^2 x^3 - k^3 x^2 - x + k$
 $k \neq 0$

$$P(x) = k^2 x^2 (x-k) - (x-k) \\ = (x-k)(k^2 x^2 - 1) \quad (1)$$

$$\therefore P(k) = 0$$

$$\text{or } k^3 - k^3 - k + k = 0$$

$$\therefore (x-k) \text{ is a factor} \quad (1)$$

$$\text{ii) } (x-k)(k^2 x^2 - 1) \quad (1) \\ = (x-k)(kx-1)(kx+1) \quad (1)$$

$$\text{for } P(x) = 0 \quad x = k \text{ or } \pm \frac{1}{k}$$

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Q 10

a) ${}^9C_4 = 126$ ①

ii) ${}^9C_4 - {}^6C_2 \times {}^3C_2$ ①

$$= 126 - 45$$

$$= 81$$
 ①

b) $2x^3 + 6x^2 + 4x + 1 = 0$

$\therefore (1-\alpha) + (1-\beta) + (1-\gamma)$

$$= 3 - (\alpha + \beta + \gamma)$$

$$= 3 - \left(-\frac{6}{2}\right)$$

$$= 6$$
 ①

ii) given that $\left. \begin{matrix} 1-\alpha \\ 1-\beta \\ 1-\gamma \end{matrix} \right\}$ are roots

$\therefore 2x^3 + 6x^2 + 4x + 1 = 0$

is $2(1-x)^3 + 6(1-x)^2 + 4(1-x) + 1 = 0$

gives a constant of 13
with leading co-efficient of -2

$$\therefore \alpha\beta\gamma = -\frac{d}{a} = -\left(\frac{13}{-2}\right)$$

$$= 6\frac{1}{2}$$

c) $\cos x - 3\sin x + 3$

$$\frac{1-t^2}{1+t^2} - 3\frac{2t}{1+t^2} + 3$$
 ①

$$\frac{1-t^2-6t+3+3t^2}{1+t^2}$$
 ①

$$= \frac{2t^2-6t+4}{1+t^2}$$

$$= \frac{2(t^2-3t+2)}{1+t^2}$$

$$= \frac{2(t-2)(t-1)}{1+t^2}$$
 ①

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Q11 contd.

Q11 a

i)

$$(1+x)^n$$

$$= {}^n C_0 + {}^n C_1 x + {}^n C_2 x^2 + {}^n C_3 x^3 + \dots + {}^n C_n x^n$$

①

ii/ LHS let $x=1$
 $\therefore (1+1)^n = 2^n$

RHS ${}^n C_0 + {}^n C_1 (1) + {}^n C_2 (1)^2 + {}^n C_3 (1)^3$
 $+ \dots + {}^n C_n (1)^n$

$$= {}^n C_0 + {}^n C_1 + {}^n C_2 + \dots + {}^n C_n$$

Question 12

a) $P(x) = x^4 + 4x^3 + 7x^2 + 6x + 2$

$$P(-1) = 0$$

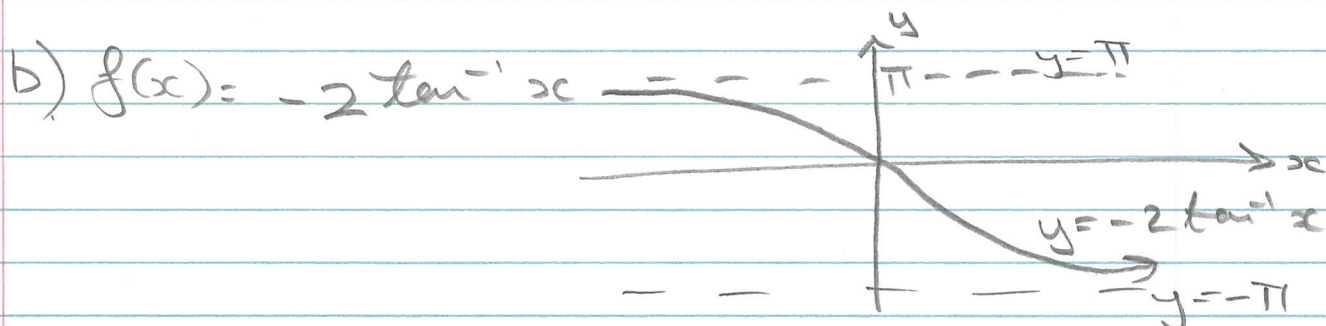
$$P'(x) = 4x^3 + 12x^2 + 14x + 6$$

$$P'(-1) = 0$$

$$P''(x) = 12x^2 + 24x + 14$$

$$P''(-1) \neq 0$$

$\therefore -1$ has a multiplicity of 2 for $P(x)=0$



c) $\sin^2 \theta = \cos \theta + 1$
 $1 - \cos^2 \theta = \cos \theta + 1$
 $\cos^2 \theta + \cos \theta = 0$
 $\cos \theta (\cos \theta + 1) = 0$

$\cos \theta = 0$ or $\cos \theta = -1$
 $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$ or $\theta = \pi, 2\pi$

Check $\frac{\pi}{2}, \frac{3\pi}{2}$ ~~and~~ $\therefore \theta = \pi$

6/8

Q12d) $x^2 + y^2 - 6x + 2y + 6 = 0$

$$x^2 - 6x + \underline{\quad} + y^2 + 2y + \underline{\quad} = -6$$

$$x^2 - 6x + 9 + y^2 + 2y + 1 = -6 + 9 + 1$$
$$(x-3)^2 + (y+1)^2 = 4$$

centre $(3, -1)$

radius 2

12 e) $f(x) = (x-2)^2$

$$f(x) = y = (x-2)^2$$

$$f^{-1}(x) \cdot xc = (y-2)^2$$
$$\pm \sqrt{x} = y - 2$$

$$2 \pm \sqrt{x} = y$$

$$y = 2 \pm \sqrt{x}$$

but for $f(x)$ $0 \leq y \leq 4$
 $0 \leq x \leq 2$

$$y = 2 - \sqrt{x}$$

$$D: [0, 4]$$

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Q 13

a) TRIANGLE

i) $\boxed{1 \text{ AE}} \leftrightarrow \boxed{\text{TRNGL}}$

$$2 \times 3! \times 5! = 1440$$

ii) $\begin{matrix} 1^{\text{st}} \& \text{last} & 5 \times 4 \\ \text{rest arranged} & 6! \end{matrix} \Rightarrow 5 \times 4 \times 6! = 14400$

b) $\sec x + \tan x = k$

$$(\sec x + \tan x)(\sec x - \tan x) = \sec^2 x - \tan^2 x = 1$$

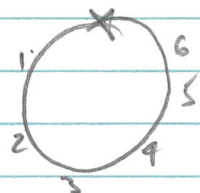
$$\therefore \sec x - \tan x = \frac{1}{\sec x + \tan x}$$

$$\text{so } \sec x + \tan x = k$$

$$\sec x - \tan x = \frac{1}{k}$$

$$2 \sec x = k + \frac{1}{k}$$

$$\sec x = \frac{1}{2} \left(k + \frac{1}{k} \right)$$

c) i)  1 fixed $\therefore 6! = 720$

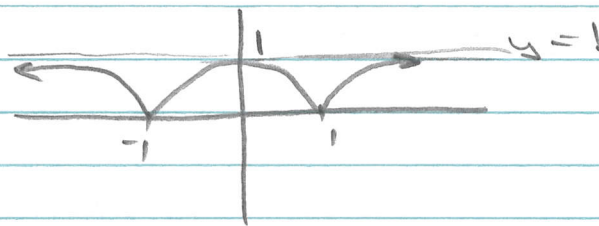
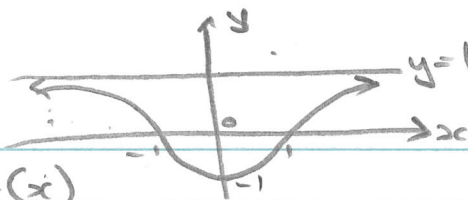
ii) Fix one girl $\therefore 2!$ ways for boys
 $5!$ for 4 girls and 1 boy

$$\text{Total } 5! \times 2! = 240$$

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d) from

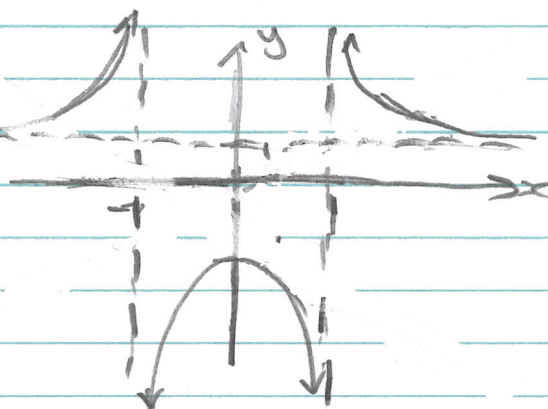
13 i) $y = |f(x)|$



① for +ve

① for shape

iii) $y = \frac{1}{f(x)}$



① for Asymptotes

① for shape

e) $\cos x - \sqrt{3} \sin x$

$$R \cos(x + \alpha) \equiv \cos x - \sqrt{3} \sin x$$

$$R \cos \alpha = 1$$

$$R \sin \alpha = \sqrt{3}$$

$$\therefore \tan \alpha = \sqrt{3}$$

$$\alpha = \frac{\pi}{3}$$

$$2 R^2 = 1 + 3$$

$$R^2 = 4$$

$$R = 2$$

for $R > 0$

$$\therefore 2 \cos\left(x + \frac{\pi}{3}\right)$$